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Herding Behavior and Investor Sentiment in an Emerging Market: Evidence from the Brazilian VIX IBOVESPA Index

Abstract
This paper investigates the relationship between herding behavior and investor sentiment in the Brazilian financial market. We employ the cross-sectional absolute deviation (CSAD) framework combined with quantile regression and use the S&P/B3 Ibovespa VIX as a proxy for investor sentiment and market uncertainty. Our sample comprises daily returns of 380 firms included in the Ibovespa index from July 1, 2021, to June 30, 2024. The results reveal no systematic evidence of herding in the Brazilian equity market, while investor sentiment exhibits a positive association with market behavior across quantiles. These findings suggest that market participants predominantly act autonomously, relying on individual assessments rather than following collective sentiment. However, during periods of heightened uncertainty associated with the COVID-19 pandemic, we observe reduced return dispersion, indicating episodic herding behavior. Even in such periods, investors appear to balance risk and return considerations, avoiding excessive reactions that could amplify volatility and destabilize market sentiment. This study contributes to the literature by exploiting the recent introduction of the S&P/B3 Ibovespa VIX, enabling an original examination of the link between market uncertainty and herding in an emerging market setting. From a practical perspective, our findings offer valuable insights for investors and policymakers seeking to monitor behavioral risks and anticipate episodes of collective market behavior.

Key words: Herd behavior; Investor sentiment; VIX; Brazilian financial market.
Paper type: Research paper.

1. Introduction

Behavioral finance focuses on incorporating the behavior of market participants into financial market analysis. It seeks to understand how psychological biases and emotional states influence market dynamics. This includes considering measures of stock market volatility, especially implied volatility, a forward-looking measure embedded in option prices that reflects expectations of future price fluctuations (Äijö, 2008; Ben-Rephael *et al*., 2012).

(Implied) Volatility often signals financial stress, rising after periods of market uncertainty, such as economic and political shocks (which may temporarily double) (Bloom, 2009). Low volatility represents financial stability (Danielsson *et al*., 2018), while high levels of implied volatility are linked to market turbulence (Magner *et al*., 2021). However, in prolonged periods of low volatility, economic agents may take excessive risks, leading to instability, such as crises (Danielsson *et al*., 2018).

Recent studies suggest the increase in implied volatility can be impacted by market/investor sentiment, playing a role in shaping perceived uncertainty. Gao *et al*. (2022) highlight that individual investors tend to exhibit irrational behaviors influenced by sentiment, which affects market stability. Agarwal *et al*. (2025) demonstrate how investors' emotional dynamics, like excitement and anxiety, are correlated with returns and volatility in different directions.

Increased implied volatility represents expectations of greater uncertainty and risk, leading investors to make similar financial decisions (Magner *et al*., 2021). This market behavior, resulting from shared sentiment among various participants, can result in herd behavior. Herd behavior is understood as replication/imitation behavior; that is, a tendency to do the same as other market participants. Thus, they take risks without adequate information or a clear understanding of the risk-return relationship. And, when faced with problems, they

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retreat in masse (Bikhchandani *et al*., 1992; Bikhchandani and Sharma, 2000; Chang *et al*., 2000).

Among the models for identifying herd behavior, the one proposed by Chang *et al*. (2000) stands out: the Cross-Sectional Absolute Deviation of Returns (CSAD). This model captures a nonlinear relationship between CSAD and market return and, thus, highlights herd behavior. CSAD was applied in several scenarios and periods, with varying results. While some studies identify herding behavior (Banerjee *et al*., 2017; Choi and Yoon, 2020; Pochea *et al*., 2017; Sheikh *et al*., 2025), others failed to demonstrate the presence of this behavior (Economou *et al*., 2018; Tlili *et al*., 2023) or captured reverse herding behavior (Ali *et al*., 2023; Choi and Yoon, 2020; Sheikh *et al*., 2025).

This behavior has been analyzed in different contexts or market participants, such as among (macro) forecasters (Clements, 2018) and institutional investors (Gavriilidis *et al*., 2024), regarding the level of transparency of company information (Wang and Huang, 2019), in mutual fund (Ben-Rephael *et al*., 2012), private equity funds (Buchner *et al*., 2020) or hedge funds (Xing *et al*., 2024), during financial crises and bubbles (Agarwal *et al*., 2025; Danielsson *et al*., 2018) and health and political crises (Assoe *et al*., 2025). In addition to different markets, regional groups, economic blocs, or economic classifications (developed and emerging markets) (Cakan *et al*., 2019; Sheikh *et al*., 2025; Shrotryia and Kalra, 2022; Vieito *et al.*, 2024).

Among them, the emerging markets are considered more fertile ground for exaggerated reactions and chain reactions, such as herd behavior (Bogdan *et al.,* 2022; Cakan *et al*., 2019; Loang, 2025; Zouaoui *et al*., 2011). From this perspective, Brazil, one of the largest emerging markets and the largest in Latin America, with significant international trade relations, provides an excellent field of study for the topic. Thereby, advancing in emerging market behavioral relationships, we aim to investigate the relationship between herd behavior and investor sentiment in the Brazilian financial market.

As a sentiment proxy, we use the Volatility Index (VIX), which offers a more exogenous perspective on market uncertainty than measures of past return volatility because it demonstrates the market's expectations or perceived risk in the face of uncertainty shocks (Äijö, 2008; Ben-Rephael *et al*., 2012; Bloom, 2009; Magner *et al*., 2021). The VIX is widely recognized and applied globally (Akin and Akin, 2024; Zhang and Giouvris, 2022). In March 2024, this index was launched for the Brazilian market, S&P/B3 Ibovespa VIX, based on the CBOE Volatility Index methodology.

As the VIX for the Brazilian market did not exist until recently, it was not known whether there was a relationship between market uncertainty, captured by the spreads between puts and calls, and herd behavior. Its inclusion fills an important gap in the literature by allowing researchers to capture domestic implied volatility, thus improving the identification of local uncertainty and potential behavioral effects. Therefore, the availability of this data makes it possible to contribute to the literature and to analyze the existence of a previously unexplored relationship between market uncertainty and herd behavior in the Brazilian scenario.

Considering the availability of the VIX Ibovespa, we analyzed the period from July 1, 2021, to June 30, 2024. For data analysis, we used the quantile regression model, which is suitable for capturing heterogeneities throughout the conditional distribution of the dependent variable, allowing the analysis of different quantiles, including at extreme points. In addition, the sample interval covers part of the COVID-19 pandemic context, a period characterized by high market volatility, ensuring sufficient variance to be captured in quantile analysis.

In summary, the results do not indicate evidence of herd behavior, and the estimated coefficients for the sentiment variable showed a positive sign, suggesting that participants in the Brazilian financial market tend to act autonomously, based on their perceptions, rather than simply following the aggregate market behavior, even in times of uncertainty (when there is an interaction between sentiment and the pandemic). However, analyzing just the pandemic

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period, we found that the dispersion of returns (CSAD) was relatively lower. This pandemic period introduced a structural break in the Brazilian market, producing temporary clustering in the returns, but not reversing the anti-herding observed in the sample overall.

The S&P/B3 Ibovespa VIX is a recent measure for the Brazilian market, which may not have captured market sentiment, especially during the COVID-19 pandemic, as it was determined by estimates. However, the study is not unfeasible, since the proposal is to include an internationally standardized uncertainty measure in analyses of the Brazilian market scenario and sentiment, and its possible impact on Brazilian market trading. In addition to proposing an analysis with a measure of domestic uncertainty, it represents the sentiment perceived internally by participants in the market. The results represent a relevant practical contribution, as they provide market participants and policymakers with a useful tool to understand and monitor behavioral risks and anticipate episodes of collective behavior.

Before the availability of the S&P/B3 Ibovespa VIX, the IVol-Br volatility measure (calculated by the Federal University of São Paulo - USP) was used. However, this data is only available until April 2022, which makes more recent analyses unfeasible. Two future study suggestions are made: firstly, that studies analyzing the pandemic period more intensively and/or exclusively include the IVol-Br measure in their analyses to capture the effect/sentiment of that moment; secondly, that they empirically compare an internal and an external measure of sentiment, such as the S&P/B3 Ibovespa VIX and the US VIX.

Finally, the S&P/B3 Ibovespa VIX proves to be a useful tool for monitoring and capturing the internal sentiment of the Brazilian market, similarly to the role played by the US VIX and analogous indices in other countries. Its use advances the literature by offering a forward-looking and standardized proxy for uncertainty specific to Brazil, filling an important gap in measuring sentiment in emerging markets, especially in Brazil. Furthermore, its application proves relevant for herd behavior analysis, allowing investigation into how domestic uncertainty affects return dispersion and investor behavioral dynamics.

The paper is organized as follows: Section 2 contains a brief definition of the concepts and previous studies on the market and herd behavior; Section 3 discusses the methodological procedures, presenting the data and the way of measuring the results; Section 4 presents the data analysis and discussion of the findings; and Section 5 summarizes the work, highlighting the main findings of the work.

2. Literature review

2.1 Herding behavior

Herding behavior refers to a replication behavior in which investors in the financial market tend to replicate the actions of other market participants, ignoring their own information (Bikhchandani *et al*., 1992; Chang *et al*., 2000).

The most common model for identifying herding behavior is the Cross-Sectional Absolute Deviation of Returns (CSAD), proposed by Chang *et al*. (2000), a simple and intuitive measure of herding, which tends to decrease in the presence of herding behavior, and is defined as follows:

$$CSAD_t = \frac{1}{N}\sum_{i=1}^{n} |R_{i,t} - R_{m,t}| \qquad (1)$$

where $R_{i,t}$ is the return of asset *i* at time *t* and $R_{m,t}$ is the average of $R_{i,t}$.

It is important to mention that CSAD is not a measure of herding. Herding behavior is detected by the relationship between $CSAD_t$ and $R_{m,t}$. When there is a linear relationship under normal market conditions, CSAD increases proportionally to the market return, indicating the absence of herding behavior (Chang *et al*., 2000; Christie and Huang, 1995).

Furthermore, if market participants are more likely to herd (follow the market) in times of crisis (market stress), herding behavior can be captured by a nonlinear relationship between CSAD and market return, expressed by:

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$$CSAD_t = \alpha + \beta_1 |R_{m,t}| + \beta_2 R_{m,t}^2 + \varepsilon_t \quad (2)$$

β_2 captures herding behavior, and the suggested interpretation of it is that if it is negative and significant, there is herding behavior (i.e., absolute market returns are high, investors ignore their private opinions and follow the market, attenuating the dispersion of returns and resulting in a decline). On the other hand, if β_2 is positive and significant, there is a reverse herd or anti-herd behavior (i.e., investors ignore the information that has been provided according to market sentiment and price movement and follow their individual opinions) (Choi and Yoon, 2020; Gebka and Wohar, 2013).

Herding behavior has already been identified in the Indian futures market (Banerjee *et al*., 2017), in Central and Eastern European countries (Pochea *et al*., 2017), in the Korean stock market (Choi and Yoon, 2020), in the Chinese market (Sheikh *et al*., 2025) or even in commodity sector energy and metals (Youssef and Mokni, 2021). Some studies have not identified the presence of this behavior, such as in Economou *et al*. (2018), in the analysis of the markets of the USA, United Kingdom, and Germany; and in Tlili *et al*. (2023), in the markets of Egypt, Jordan, Morocco, and Tunisia.

It is particularly relevant to analyze emerging markets, as they are considered markets more prone to irrational/behavioral biases, such as herd behavior (Bogdan *et al.,* 2022; Loang, 2025). Vieito *et al.* (2024) analyzed the Latin American Integrated Market (MILA), composed of the stock exchanges of Chile, Peru, Colombia, and Mexico, and found significant herd behavior in all markets. In turn, Sheikh *et al.* (2025) analyzed the Chinese and Pakistani markets, finding herd behavior in the Chinese market and reverse herd behavior in the Pakistani market.

Zhang and Giouvris (2022) analyzed the presence of herding for the economic bloc of emerging markets, the BRICS, and found that herd behavior is present only in China (during the global financial crisis). Analyzing this same economic bloc, Shrotryia and Kalra (2022) identified herd behavior in China and South Africa but did not find it in the Brazilian, Indian, and Russian stock markets. On the other hand, Loang (2025) identified the presence of herd behavior in all BRICS countries.

In the context of emerging markets, Brazil stands out as one of the largest emerging markets and the largest in Latin America, making it a relevant environment for studies such as this one. In terms of Brazilian studies, Zulian *et al*. (2012) analyzed Brazilian equity investment funds and identified the existence of herding behavior with an intensity similar to that of countries such as the United States, the United Kingdom, and Germany. In the Brazilian stock market, Signorelli *et al*. (2021) analyzed the period from 2008 to 2019 and identified herd behavior in several years throughout this interval.

In light of the above, we present the first research hypothesis of this study:

H1: Herd behavior is present in the Brazilian stock market.

The discussion of herd behavior also involves understanding which additional factors can influence it. In this sense, space is opened for the discussion of the relationship between investor (or market) sentiment and herd behavior. Among the ways to identify and measure this sentiment is the use of the Volatility Index (VIX), presented below.

2.2 Volatility Index (VIX)

Implied volatility reflects expectations of future fluctuations in option prices (Äijö, 2008; Ben-Rephael *et al*., 2012), widely used to represent financial stress, characterizing the optimism or pessimism of the market. In synthesis, low volatility represents financial stability (optimism), and high levels are link associates ed to market turbulence (pessimism) (Bloom, 2009; Danielsson *et al*., 2018; Magner *et al*., 2021).

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The CBOE Volatility Index (VIX) reflects the market's collective estimate of the level of future uncertainty, but it is not a tradable product. Unlike realized volatility, which measures historical price variation, the VIX indicates the volatility expected by the market (implied volatility), that is, the perceptions of market participants regarding future scenarios. In this sense, VIX values are calculated by weighting the prices of the S&P 500 (SPX) call and put options (rather than stocks) for the next 30 days.

VIX values can be classified as: "Low" (0-15) generally indicating market optimism; "Moderate" (>15-20), normal market environment; "Medium" (>20-25), increasing market concern; "High" (>25-30), market turbulence; and "Very High" (>30+) extreme market turbulence.

Magner et al. (2021) associate elevated VIX levels with periods of market turbulence and argue that the index influences market returns and reflects global economic uncertainty, with significant implications for investors and policymakers. In times of heightened uncertainty, investors tend to follow collective decisions, making the VIX a proxy for measuring the relationship between uncertainty and investor behavior. Likewise, Youssef and Mokni (2021) suggest that herding behavior is stronger during periods of higher volatility.

Economou *et al*. (2018) documented the statistically significant and negative impact of VIX on herding estimations of the USA, UK, and Germany, having adopted in their analyses the specific VIX of each of the markets, all based on the CBOE methodology. Choi and Yoon (2020) analyzed the connection between herd behavior and investor sentiment, adopted the volatility index of the Korean market (VKOSPI), and observed a positive relationship between internal sentiment and (reverse) herd behavior in the KOSDAQ market.

Zhang and Giouvris (2022) analyzed the relationship between American VIX and the markets of the countries that make up the BRICS, and found that these markets, especially Brazil, are negatively affected by the VIX (of the US), regardless of periods of crisis or not. Furthermore, Akin and Akin (2024) identified a negative and significant relationship between the VIX and the S&P 500.

Returning to the argument about the greater propensity for herd behavior on the part of emerging countries (Bogdan *et al.,* 2022), specifically Brazil, it was identified that in March 2024, the S&P/B3 Ibovespa VIX was launched for the Brazilian market, and based on the CBOE Volatility Index methodology, reflecting perceived risk through implied volatility in Ibovespa B3 option prices. It is pertinent to mention that this methodology was only possible to apply in Brazil due to the growing liquidity in Brazil's options market. As it is an index recently launched for the country's financial market, no previous studies were identified using the S&P/B3 Ibovespa VIX. The previous relationship between VIX and the Brazilian market was not based on domestic indicators but on external indicators (Zhang and Giouvris, 2022).

Therefore, given that an increase in the VIX can indicate periods of uncertainty and pessimism and generate a chain reaction (herd) behavior, the second hypothesis of this study is:

H2: Brazilian market sentiment (VIX) negatively influences herd behavior (CSAD).

3. Methodology

3.1 Data

To attend to the proposed objective, the daily CSAD (dependent variable) was obtained following the methodology of Chang *et al*. (2000), as described in equation 1. The daily returns of all companies active in the Brazilian market (B3 – Brasil, Bolsa, Balcão) ($R_{i,t}$) were collected from the Economática database, as well as the daily return of the Ibovespa Index ($R_{m,t}$). To

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represent investor sentiment, the S&P/B3 Ibovespa VIX (Sent) was considered, which was collected from the S&P Dow Jones Indices website.

It is worth noting that the VIX for the Brazilian market was launched in March 2024; the indices prior to that date are the result of retroactive estimates (starting in May 2021) calculated and provided by S&P Dow Jones Indices itself and based on the CBOE Volatility Index methodology. Due to this disclosure, the period from July 1, 2021, to June 30, 2024, was considered for analysis. Thus, the sample consists of 747 daily returns of 380 companies listed on the Brazilian stock market and the respective daily VIX of the Brazilian market.

Figure 1A shows the relationship between $CSAD_t$ and $R_{m,t}$ in the period studied. It is possible to perceive a grouping of the data instead of a linear increase/decrease. Furthermore, it is suggested that the data present reverse herding behavior since the deviations of the returns go upwards when at the points extremes of the market.

The time series of the variables CSAD and *Sent* (represented by the VIX) are reported in Figure 1B and Figure 1C, respectively. Based on the findings, supported by the description in Table I, it is possible to see the minimum CSAD value close to 1 in the final period of the series (first half of 2024), indicating a low dispersion among the returns. The low dispersions suggest (but do not guarantee) the presence of herd behavior. The maximum value for CSAD was above 3.5 in the second half of 2022, which may indicate that there is greater dispersion among the returns and investors are not following the market. Indicating, therefore, there would be no herd behavior in the sample.

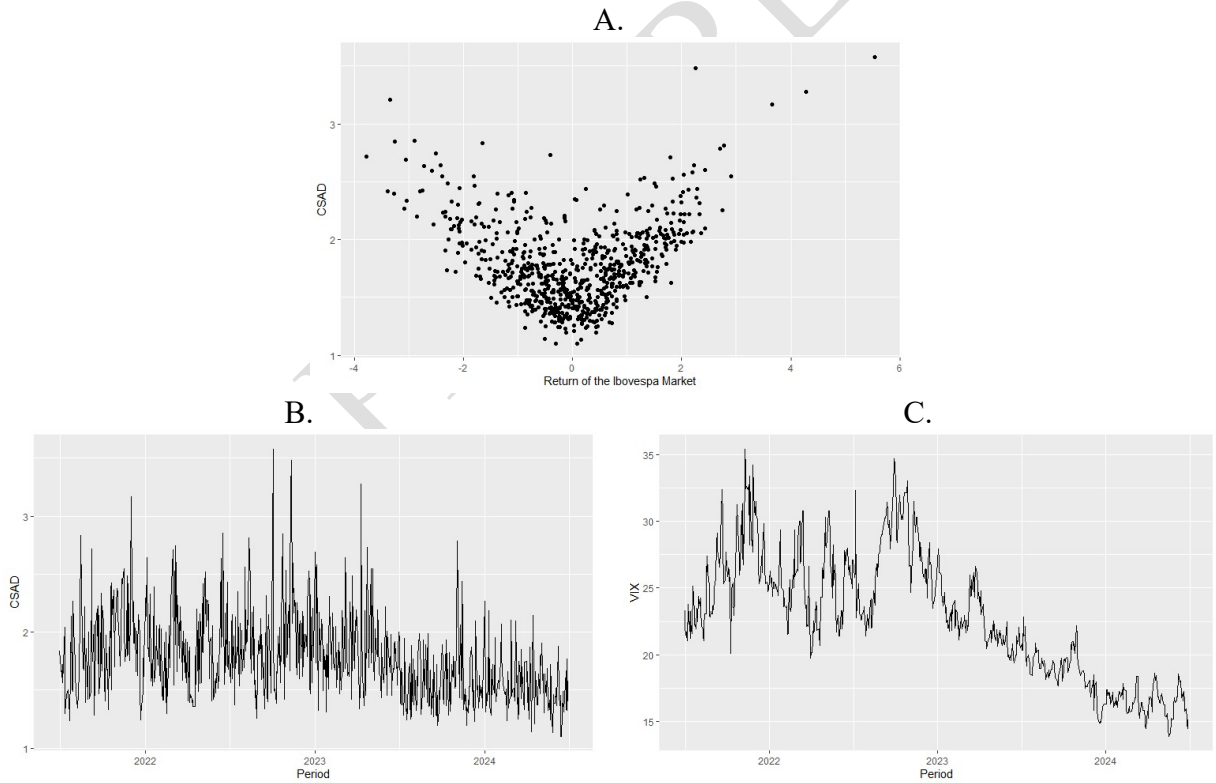

Figure 1 A: relationship between $CSAD_t$ and $R_{m,t}$; B: time series CSAD; and C: time series *Sent* (VIX).
Source: Authors own work

It is worth noting that there are no scales for CSAD; that is, the minimum and maximum values to indicate herd behavior are not pre-established. Therefore, these values refer to the sample and are possible indications of results. However, they do not guarantee a conclusion by themselves.

Regarding the Sent variable, the minimum value presented for the VIX is 13.89 (in 2024), which is classified as "Low" (optimistic market). The average and median are close, remaining in the "Average" classification. In turn, the maximum value presented was above 30 and classified as "Very High".

All indexes greater than 30 were found in the years 2021 (on at least one day in September to December) and 2022 (on at least one day in March, May, July, September, October, and November), and may be a reflection remnant of the COVID-19 pandemic outbreak, as well as the political uncertainties of the election year or even extern factors. However, it is worth noting that the VIX for the Brazilian market was launched in March 2024, and indexes prior to that date are the result of retroactive estimates.

In this sense, as the pandemic period was quite volatile and could influence the sample (Ghorbel *et al*., 2023; Zhang and Giouvris, 2022), we include a robustness check with a COVID dummy to test whether the relationship between implied volatility (Sent) and Cross-Sectional Return Dispersion (CSAD) changes during the pandemic. Therefore, the pandemic period is defined as July 1, 2021, to May 22, 2022 (*Covid* = 1), and the post-pandemic period is defined as May 23 to June 30, 2024 (*Covid* = 0). Thus, according to Table I, approximately 30% of the sample refers to the pandemic period.

	Minimum	**Maximum**	**Mean**	**Median**	**Standard deviation**	**DF**	**DFA**
CSAD	1.101	3.577	1.771	1.713	0.357	-1.928 §	-14.163 ***
$\lvert R_{m.t}\rvert$	0.001	5.541	0.917	0.740	0.737	-7.863 ***	-19.789 ***
$R^2_{m.t}$	0.000	30.708	1.384	0.548	2.290	-13.058 ***	-19.627 ***
Sent	13.890	35.390	22.760	22.530	4.654	-0.814	-4.977 ***
Covid	0.000	1.000	0.296	0.000	0.457	-1.842 §	-1.603

Table I – Descriptive statistics and unit root tests
Note: Statistical significance: 0 '***' 0.001 '**' 0.01 '*' 0.05 '§' 0.1 ' ' 1
Source: Authors own work

Furthermore, to test the stationarity of the time series, the Dickey-Fuller (DF) and Augmented Dickey-Fuller (DFA) tests were performed, with the inclusion of trends and a constant; the test statistics are reported in Table I. All DFA statistics were significant, and it can be demonstrated that the series does not have a unit root. Therefore, they are stationary.

3.2 Measurement

The CSAD value is commonly used in a nonlinear regression model, as described in Equation 2. On the other hand, we propose a more detailed analysis using quantile regression, which accounts for data beyond the mean and is especially effective at capturing heterogeneities in the conditional distribution of the dependent variable (CSAD). Another factor is that the sample interval covers part of the COVID-19 pandemic context, a period characterized by high market volatility, ensuring sufficient variance to be captured in quantile analysis. In addition, we include in the model investor sentiment measured by the S&P/B3 Ibovespa VIX and control for the pandemic period.

We consider the use of the quantile regression (QR) model proposed by Koenker and Bassett (1978) and used in previous studies investigating herding behavior (Choi and Yoon, 2020; Pochea *et al*., 2017; Sheikh *et al*., 2025; Shrotryia and Kalra, 2022; Stavroyiannis and Babalos, 2017; Tlili *et al*., 2023), exploring the most extreme points of the distribution of the dependent variable. Thus, an adaptation to Equation 2 is defined as follows:

$$CSAD_t(\tau) = \alpha(\tau) + \beta_1(\tau)\left|R_{m,t}\right| + \beta_2(\tau)R^2_{m,t} \qquad (3)$$

where τ represents the different quantiles that will be analyzed: 10, 30, 50, 70, and 90.

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Still meeting the objective of the study, the variable *Sent* is included in the regression model, and the following relationship is analyzed:

$$CSAD_t(\tau) = \alpha(\tau) + \beta_1(\tau)\left|R_{m,t}\right| + \beta_2(\tau)R_{m,t}^2 + \beta_3(\tau)Sent_t \quad (4)$$

where τ represents the quantiles: 10, 30, 50, 70, and 90.

The period analyzed was marked by high uncertainty due to the pandemic crisis, resulting in high volatility and significant variance in the data, a scenario conducive to the application of quantile regressions. To further investigate the effects associated with the period of coping with the COVID-19 pandemic, a dummy variable was incorporated to control for the difference between the pandemic and post-pandemic periods. In addition, we included the interaction between this dummy variable and the $Sent_t$ variable to verify possible changes in the impact of sentiment over time. Thus, the model can be specified as follows:

$$CSAD_t(\tau) = \alpha(\tau) + \beta_1(\tau)\left|R_{m,t}\right| + \beta_2(\tau)R_{m,t}^2 + \beta_3(\tau)Sent_t + \beta_4(\tau)Covid_t + \beta_5(\tau)Sent_t * Covid_t \quad (5)$$

where τ represents the quantiles: 10, 30, 50, 70, and 90; and $Covid_t$ = 1 is a dummy variable to represent the pandemic period.

4. Results

4.1 Herding behavior

To analyze the CSAD distribution in more detail, Table II (Panel A) presents the results estimated using Equation 3. The coefficient β_1 was positive and significant in all quantiles measured, a similar result to that obtained by Ghorbel *et al*. (2023), Loang (2025) and Sheikh *et al*. (2025). This finding suggests that on days of extreme price movements, the dispersion of returns (CSAD) increases. Therefore, it follows that, in periods of market stress, investors are dispersing and not following the market (Chang *et al*., 2000).

Table II (Panel A) shows positive β_2 coefficients in all quantiles, with significance only at quantiles 0.10 (τ=0.10) and 0.70 (τ=0.70), which suggests the presence of reverse herd behavior (or anti-herd). In anti-herd behavior, noisy/irrational investors ignore market movements and the information provided (based on market sentiment and price movements), and trade based on inaccurate information, neglecting fundamental values and increasing the dispersion of returns with increasing rates (Choi and Yoon, 2020; Sheikh *et al*., 2025). Similar results were identified in the Pakistani market (Sheikh *et al*., 2025), in the Korean stock market (Choi and Yoon, 2020), in US hedge funds (Ali *et al*., 2023), and even in Brazil (Shrotryia and Kalra, 2022).

A possible explanation for our results is that emerging markets, such as Brazil, tend to exhibit high investor heterogeneity (individual or retail investors, foreign participants, and institutional traders) and greater informational asymmetry, resulting in fragmented decision-making. These characteristics reduce the likelihood of coordinated (herd) behavior. The Brazilian market is mostly composed of individual investors, who may be considered inexperienced or noisy; this finding may indicate that, during the analyzed period, they ended up reacting on their own and following the path that seemed most appropriate to them, acting in anti-herd.

Furthermore, considering that herd behavior is identified daily through the dispersion of returns, it can be assumed that the presence of herd behavior was diluted over the analyzed period. Therefore, to strengthen the findings, a more detailed analysis of the uncertainty period present in the sample (effects of COVID-19) is presented in section 4.3.

The significance in the 0.10 and 0.70 quantiles provides additional insights into the dynamics of investor behavior. At the 0.10 quantile (when there is low dispersion of returns, i.e., low CSAD), asset and market returns tend to cluster, possibly due to calmer market conditions. However, the clustering of returns does not imply the clustering of decisions in the market and, consequently, herd behavior. Therefore, investors tend to follow heterogeneous

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signals instead of converging on common strategies, reinforcing the idea of fragmented decision-making mentioned earlier.

Similarly, when market movements intensify and dispersion increases (at the 0.70 quantile), investors continue to diverge in their interpretations and reactions rather than converge toward herd behavior. The presence of anti-herding at both low and high CSAD levels highlights the structural heterogeneity of the Brazilian market, where behavioral responses remain fragmented across different market conditions.

Panel A: $CSAD_t(\tau) = \alpha(\tau) + \beta_1(\tau)\lvert R_{m,t}\rvert + \beta_2(\tau)R_{m,t}^2$				
	α	**β1**	**β2**	**AIC**
τ = 0.10	1.228 ***	0.242 ***	0.033 ***	52.279
τ = 0.30	1.320 ***	0.291 ***	0.021	-28.913
τ = 0.50	1.455 ***	0.273 ***	0.025	39.184
τ = 0.70	1.585 ***	0.232 ***	0.048 *	206.824
τ = 0.90	1.802 ***	0.279 **	0.031	636.323

Panel B: $CSAD_t(\tau) = \alpha(\tau) + \beta_1(\tau)\lvert R_{m,t}\rvert + \beta_2(\tau)R_{m,t}^2 + \beta_3(\tau)Sent_t$					
	α	$\boldsymbol{\beta_1}$	$\boldsymbol{\beta_2}$	$\boldsymbol{\beta_3}$	**AIC**
τ = 0.10	0.996 ***	0.218 ***	0.033 ***	0.012 ***	-17.984
τ = 0.30	1.020 ***	0.240 ***	0.023 *	0.016 ***	-141.482
τ = 0.50	1.037 ***	0.251 ***	0.018	0.020 ***	-92.521
τ = 0.70	1.048 ***	0.237 ***	0.036 *	0.024 ***	83.330
τ = 0.90	1.121 ***	0.182 *	0.041 §	0.032 ***	469.803

Table II – CSAD Estimates
Note: Statistical significance: 0 '***' 0.001 '**' 0.01 '*' 0.05 '§' 0.1 ' ' 1
Source: Authors own work

4.2 Herding Behavior and Investor Sentiment

Table II (Panel B) reports the data including the *Sent* variable in the analysis, measured by the S&P/B3 Ibovespa VIX. Similar to what was mentioned previously, the coefficient associated with $R_{m,t}^2$ was positive and significant in all quantiles measured, at least at 10%, except at 0.50. The finding points to anti-herd behavior in the Brazilian financial scenario. Thus, it suggests that, in the Brazilian financial market, investors do not imitate the behavior of other participants but follow their own perceptions and ignore market information, increasing dispersion (CSAD).

About the variable of interest (*Sent*), it can be seen that all estimates in β_3 are positive and significant in all quantiles, indicating that when the VIX of the Brazilian market increases (i.e., uncertainty increases), CSAD also increases. This positive relationship was also evidenced in other emerging markets such as Banerjee *et al*. (2017), Choi and Yoon (2020), and Sheikh *et al*. (2025). Therefore, it is suggested for the present sample that in the face of market uncertainty, investors in the Brazilian market do not gather and do not act in a herd.

The Brazilian diversified market generates a fragmented decision-making environment. As implied volatility rises, the different market participants interpret risk differently, leading to greater return dispersion. Thus, the positive Sent coefficient reflects not market efficiency, but supports the notion that uncertainty in emerging markets often triggers idiosyncratic responses rather than collective action.

Our findings contrast with the results of Zhang and Giouvris (2022) regarding the relationship between the VIX and the Brazilian market. It is worth noting, however, that the authors analyzed a sample composed of several emerging markets and used the VIX of the United States as an indicator of uncertainty. In this sense, our research advances by offering an analysis focused exclusively on the Brazilian market, considering domestic sentiment about the market itself, evidencing a relevant contribution to the literature.

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Comparing the goodness-of-fit criteria (AIC) evidenced in Panels A and B of Table II, an improvement in the estimates is observed, given by the inclusion of the variable Sent. This suggests that the inclusion of market sentiment (VIX) is relevant to explain and analyze herd behavior in the applied sample. As implied volatility embeds forward-looking expectations derived from option prices, the findings indicate that investor sentiment plays a significant role in behavioral analysis in the Brazilian market.

In addition, the graphical analysis in Figure 2 ratifies the findings described in Table II (Panel B) and highlights the behavior of the variables in other quantiles not reported in the table. Both the absolute market return ($|R_{m,t}|$) and investor sentiment (Sent) are significant at all quantiles (since 0 is not included in the confidence interval of the estimates).

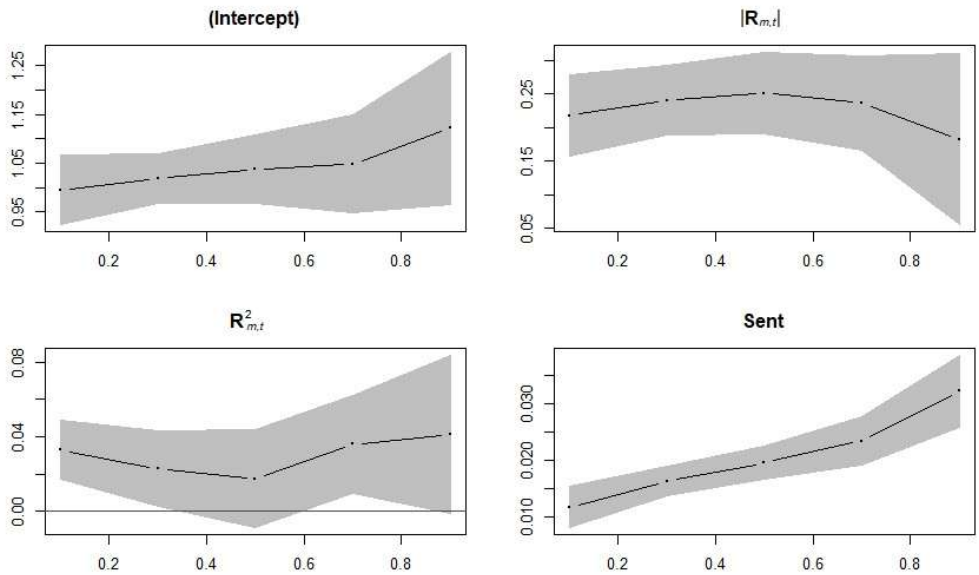

Figure 2 Graphical analysis: Equation 4
Source: Authors own work

As the CSAD quantiles increase, up to quantile 0.50, the estimated coefficients associated with $|R_{m,t}|$ increase. After quantile 0.50, a downward trend is observed, becoming more noticeable after quantile 0.70. Regarding Sent variable, an upward trend is observed in the estimates. As the CSAD quantiles increase, the coefficient estimates also become larger. Therefore, the greater the market uncertainty, the more dispersed the market participants.

Regarding $R^2_{m,t}$, Figure 2 allows us to observe the non-significance in other quantiles besides those reported. Thus, $R^2_{m,t}$ loses significance right after quantile 0.30 and becomes significant again after quantile 0.60.

4.3 Effects of COVID-19

To give greater robustness to the results, we include a control variable for the pandemic period in order to verify the effects of COVID-19 in the sample. The coefficients obtained for β_4 are negative and statistically significant in the quantiles 0.10, 0.30, and 0.50, according to Table III, indicating that the pandemic period was characterized by lower levels of return dispersion (CSAD).

This negative impact of COVID was expected (Ghorbel *et al*., 2023), and at first glance, we might be motivated to assume that the Brazilian market exhibited herd behavior. However, the coefficients in β_2 remain positive, thus evidencing anti-herding behavior. Therefore, what can be suggested is that, in a scenario of uncertainty such as a pandemic, the returns tend to

converge, but in the sample, investors interpret and make decisions on their own, without reversing the anti-herding observed in the sample overall.

During the COVID crisis, Zhang and Giouvris (2022) found a negative coefficient associated with $R^2_{m,t}$, but it was not significant. Therefore, they could not conclude that there was herd behavior in the Brazilian market. In turn, Ghorbel et al. (2023) analyzed emerging markets, including Brazil, and found herd behavior for this market. However, it can be considered that the difference exists due to the period analyzed. The authors analyze the period since the beginning of the uncertainty shock caused by the COVID-19 crisis, in December 2019, while our sample begins in July 2021. Although this period still encompasses the pandemic phase, it is possible that the market was already more adapted to the circumstances and, therefore, operated with independent strategic decision-making.

$CSAD_t(\tau) = \alpha(\tau) + \beta_1(\tau)\lvert R_{m,t}\rvert + \beta_2(\tau)R^2_{m,t} + \beta_3(\tau)Sent_t + \beta_4(\tau)Covid_t + \beta_5(\tau)Sent_t * Covid_t$							
	α	$\boldsymbol{\beta_1}$	$\boldsymbol{\beta_2}$	$\boldsymbol{\beta_3}$	$\boldsymbol{\beta_4}$	$\boldsymbol{\beta_5}$	**AIC**
τ = 0.10	1.021 ***	0.220 ***	0.033 *	0.011 ***	-0.411 **	0.016 *	-39.737
τ = 0.30	1.023 ***	0.243 ***	0.020 *	0.016 ***	-0.436 **	0.016 **	-152.487
τ = 0.50	1.049 ***	0.259 ***	0.017	0.019 ***	-0.339 §	0.013	-94.591
τ = 0.70	1.028 ***	0.232 ***	0.036 *	0.025 ***	-0.280	0.009	83.294
τ = 0.90	1.137 ***	0.189 *	0.041 §	0.031 ***	-0.099	0.004	472.822

Table III – Effects of COVID-19
Note: Statistical significance: 0 '***' 0.001 '**' 0.01 '*' 0.05 '§' 0.1 ' ' 1
Source: Authors own work

The interaction term between $Sent * Covid$ provides additional insights into how VIX operated during the pandemic. We observe positive and statistically significant coefficients in β_5 at the 0.10 and 0.30 quantiles of the CSAD, indicating that the sentiment perceived in the Brazilian market during COVID-19 continues to increase CSAD, reinforcing the argument that decisions are fragmented even during an external shock of uncertainty. Thereby, this result supports the idea that the COVID-19 and sentiment domestic operate through related channels but affect investor behavior differently in the present sample.

When the period of stress or uncertainty interacts with volatility (market sentiment), anti-herd behavior continues to occur. This may suggest greater familiarity of investors with the pandemic period (which began in 2019/2020), since the sample began in July 2021, it is plausible that, even in the face of persistent uncertainty, economic agents were already more adapted to the new scenario and the risk versus return context, which allowed them to act with greater independence and strategic basis. Or, because the VIX in the period of COVID is the result of estimates, the real sentiment could not be captured.

5. Conclusion

This study investigates the relationship between herd behavior and investor sentiment in the Brazilian financial market, using the CSAD model proposed by Chang *et al*. (2000), additionally by quantile regression to capture effects across different levels of return dispersion. As a proxy for investor sentiment, we adopted domestic implied volatility, which reflects specific market conditions, allowing us to isolate the internal component of uncertainty. That's why we studied the recently launched VIX for the Brazilian market (S&P/B3 Ibovespa VIX) and we investigated the effects in the period of coping with the COVID-19 pandemic.

Overall, the findings do provide evidence of anti-herding behavior among investors in the Brazilian financial market during the period analyzed, leading to the rejection of H1. The positive and significant coefficients associated with $R^2_{m,t}$ at several quantiles, indicate that investors tend to diverge in their reactions when market movements intensify. This finding reflects the structural heterogeneity of the Brazilian market, where retail investors, institutional

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participants and foreign investors process information differently, reducing the likelihood of aligned behavior (herding). This result persists even when we isolate the pandemic period in the sample. Although the dispersion of returns (CSAD) decreases during pandemic, the behavior remains anti-herd.

The findings indicate a positive and significant relationship between investor sentiment and CSAD, suggesting that increases in implied volatility amplify differences in individual reactions. Therefore, H2 was not supported. It is important to emphasize that the positive relationship found does not reflect market efficiency, but supports the idea of a diversified market, different strategies, and heterogeneous reactions.

Thus, regarding the variable Sent, it can be suggested that: 1) the Brazilian market managed to balance shared sentiment, not generating euphoria and replication; 2) considering that the study sample began in July 2021, it is plausible that, even in the face of persistent uncertainty of the pandemic, economic agents were already more adapted to the new scenario, which allowed them to act with greater independence and strategic foundation; 3) the VIX is still a recent indicator for the market and it has not consolidated; 4) since the VIX data before March 2024 are the result of estimates, the real sentiment could not be captured.

The domestic sentiment (*Sent*) increases dispersion of returns (*CSAD*), while uncertainty (*Covid*) decreases. When these variables are considered interacted, it is observed that the sentiment perceived in the Brazilian market during COVID continues to increase the dispersion of returns, reinforcing the argument for fragmented decisions even during a shock of uncertainty.

There is no consensus on herding behavior across different markets, and, in the case of Brazilian market, our results contrast with the negative relationship between VIX and CSAD reported by Zhang and Giouvris (2022). However, the VIX applied referred to that of a foreign market (the US). This study fills an important gap in the literature by including the domestic VIX of Brazilian market, allowing researchers and futures studies to capture domestic implied volatility, emphasizing local uncertainty, and potential behavioral effects; improving the analysis of herding behavior in emerging markets, especially in Brazil.

The use of the S&P/B3 Ibovespa VIX constitutes a significant empirical advancement because it enables, for the first time, the direct assessment of sentiment and perceived uncertainty from the Brazilian options market, rather than on foreign indicators. The main limitation of the study relates to the availability of the S&P/B3 Ibovespa VIX index, which provides a recent database and does not allow for the analysis of a long historical series. Additionally, data before its availability is found through estimates and may not represent the true sentiment of the period.

Since the S&P/B3 Ibovespa VIX is a recent measure for the Brazilian market in the period analyzed, it may be suggested that it was not yet consolidated in the market or that it did not reflect the sentiment of the time, especially during the COVID-19 pandemic, since it was determined through estimates. However, this does not preclude its application in this study since the proposal is to include in the analysis of the Brazilian market and sentiment an internationally standardized measure of uncertainty, in order for investors and policymakers to adopt this measure in the analysis of uncertainties, risks, and decision-making.

Note that before the availability of the S&P/B3 Ibovespa VIX (using the CBOE methodology), another measure of market volatility was used (IVol-Br). However, this data has been discontinued, making recent studies unfeasible. Thus, two suggestions for future studies are presented: first, that studies that aim to specifically and exclusively analyze the pandemic period may include the IVol-Br measure in their analyses to capture the effect/sentiment of that moment in addition to the VIX (calculated by the CBOE); second, studies that they empirically compare an internal and an external measure of sentiment, such as the S&P/B3 Ibovespa VIX and the US VIX, for longer series, given the availability of future data.

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